GOBI M350 Fieldbook

APPN Aerial SOP — Locked revision 1.03

APPN – GOBI M350 Fieldbook

For a high-level description of the GOBI sensor package and how it sits alongside the other APPN platforms, see the Platforms Overview.

For the GOBI fieldbook covering the Inspired Flight IF1200A, see GOBI IF1200 Fieldbook.

This fieldbook provides a standardised operational guide for APPN GOBI UAV deployments **on the DJI M350**, supporting safe flight operations, consistent sensor configuration, and high-integrity data capture. It is intended for trained APPN staff conducting hyperspectral, LiDAR, RGB, and GNSS-INS data acquisitions, and promotes repeatability, transparency, and confidence in downstream data analysis across APPN operations.

> **Important**
>
> **This protocol must be followed for all standard APPN GOBI M350 UAV flights.** Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Document Structure

1. Equipment Checklist — what to bring to the field.
2. Preflight Planning — survey / capture polygons and DJI Pilot 2 flight planning.
3. Onsite Preflight Operations — weather and airspace checks, panel and GCP layout, aircraft setup.
4. GOBI Sensor Configuration — sensor power-up, exposure, dark reference, and mission start.
5. Flight Operations — takeoff, dynamic alignment, autonomous mission, and landing.
6. Post-Flight Sensor Configuration — stopping the mission and confirming data on the sensor.
7. Data Confirmation — expected file counts and sizes by sensor.
8. Post-Flight — pack-down of aircraft, payload, and ground reference kit.
9. Offload Data — downloading from the sensor and storing under the APPN folder structure:

- Data storage, processing & validation

10. Appendix — FTP & service reference, Headwall polygon tool workflow, static IP setup, and IP address reference.

Equipment Checklist

> **Note**
>
> Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

☐ **Aircraft**
☐ DJI M350
☐ Aircraft batteries and spares
☐ Landing gear
☐ Landing pad
☐ Spare parts
☐ Tools
☐ Logbook
☐ **Radio Control Transmitter / Ground Control Station**
☐ **GRYFN Gobi**
☐ Gobi system (power once each month to charge the RGB camera internal battery)
☐ Power cables, chargers
☐ Ethernet cable (and dongle if needed)
☐ SD card
☐ *Capture* and *survey* polygon files
☐ **Ground reference kit**
☐ 2 x Reflectance calibration panels (11%, 30%, 56%, 82%)
☐ 1 x Calibration validation panels (20%, 45%)
☐ 5 × Propeller Aeropoints or other ground control points (GCPs) and/or RTK GNSS system
☐ 2 × folding tables to elevate panels
☐ *If over 30 km from a single CORS base station*, a portable RTK base station (link to GRYFN gitbook)
☐ **Accessories**
☐ Safety gear (signage and high-vis vests)
☐ Aeronautical radio
☐ Field laptop and spare batteries
☐ External storage media
☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
☐ Portable fan (for cooling GOBI during data offload)
☐ External power brick (for charging UAV RC)

Note

The **Ground reference kit** above assumes the Standard Dual ELM panel flight design. Other flight designs may require additional equipment (e.g. extra reflectance panel sets, additional GCPs and folding tables for multi-flight captures, or extra targets for validation flights). See the Standard Flight Procedure and the Validation Flight Procedure for the full equipment requirements of each design.

Preflight Planning

Important

Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance. Always use the **latest version** of the GRYFN Flight Calculator when planning flights. Further planning documentation for the GOBI can be found in the GRYFN operations gitbook.

1. Using a GPS survey system (Emlid, Trimble…) or GIS software, create a polygon with a 5 m margin around the area of interest. Make sure the polygon includes the areas where you will place your calibration panels and GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Create two polygons — a *survey* polygon and a *capture* polygon. Ensure the *survey* polygon is 2× the flight speed larger than the *capture* polygon in the direction of travel (as per the figure below). Save both polygons uniquely as KML files using the convention `YYYYMMDD_XXXX_capture` and `YYYYMMDD_XXXX_survey` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).

Note

Buffer the *capture* polygon **perpendicular to the flight direction** by at least one flight-line spacing on each side. When a flight line runs immediately adjacent to the capture-polygon boundary, the aircraft can drift out of the capture region and hyperspectral triggering stops, losing data along that edge. A perpendicular buffer provides margin for trajectory variation and ensures full capture on the outermost lines. **Increase the buffer further when flight lines cross the polygon boundary at an angle** (e.g. north–south APEx flights): the aircraft enters and exits the polygon at an angle, so a triggered frame can fall half inside and half outside the capture region.

3. *If using QGIS*, import the *capture* KML into Google Earth and immediately export it again. This is due to the required formatting Google Earth supplies, versus QGIS.
4. Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation bounds of the hyperspectral sensor.
5. Load the *survey* KML onto a USB stick or microSD card.
6. Power on the controller. Open DJI Pilot 2 and select **Flight Route → + Import Route (KMZ/KML)**. Alternatively, you can create an Area Route on the controller. *Note that you do not need the UAV powered on for this, just the controller.*
7. Using both DJI Pilot 2 and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest. See the Standard Mission Parameters table below for examples of common flight types.
8. Set the frame period to $>$ **20% oversampling (30% is our default)**. Frame period sets the along-track sampling and is separate from side overlap. Then ensure **side overlap is driven by the VNIR (hyperspectral) requirement, not the RGB camera** (see the important note under Standard Mission Parameters): for hyperspectral missions (Types 1–3) plan at **80% side overlap from the RGB custom camera**, which the GRYFN flight calculator resolves to approximately **37% VNIR side overlap**. RGB-only missions (Type 4) have no VNIR requirement and may use 75%.

Standard Mission Parameters

Standard Mission Type	Scenario	Altitude (m)	Speed (m/s)	Frame Period (Hz)	Side Overlap % (DJI)*
Type 1	Plant counting / small structures (intra-plot differences)	30	2.1	5.32 (20% oversampling)	80%
Type 2	Plant breeding experiments (inter-plot differences)	50	3.2	5.09 (30% oversampling)	80%
Type 3	Large landscape measurements (strip trials, hyperspectral transects)	80	5.1	5.11 (30% oversampling)	80%
Type 4	Surveys using LiDAR and RGB only (ecosystem measurements, forestry)	100	9	N/A	75%

* Select *custom camera* in DJI Pilot 2 flight planning and enter the **RGB camera** parameters per the GRYFN custom camera settings guide. The percentages above are the **RGB** side overlap entered into the flight planner.

Important

Overlap must be driven by the VNIR (hyperspectral) data-quality requirement — RGB overlap alone is not a sufficient measure of acceptable sidelap. Historically the standard 75% RGB setting was assumed to guarantee adequate VNIR sidelap; in practice it produces *very low* VNIR overlap and has caused missing pixels, linear data holes, and stronger edge/directional effects in recent flights.

For hyperspectral missions (Types 1–3) plan at **80% RGB side overlap**. The GRYFN flight calculator resolves this to approximately **37% VNIR side overlap** at these altitudes. VNIR side-overlap targets:

- **~37%** — delivered by the 80% RGB standard; treat as the practical minimum.
- **≥ 40%** — preferred, for robustness under wind and trajectory variation.
- **≥ 50%** — every ground point is imaged by **at least two flight lines**, greatly increasing redundancy and resilience to trajectory error. Increase RGB overlap further where this level of redundancy is required.

RGB frame cameras remain the limiting factor for the orthomosaic — GRYFN recommends an absolute minimum of 70% overlap/sidelap, with 80% giving better feature matching — so planning at 80% RGB keeps RGB products well within spec while lifting VNIR sidelap to an acceptable level. **RGB-only missions (Type 4)** have no VNIR requirement and may use 75%.

Note

Frame-period values may differ slightly between the **GRYFN Flight Calculator** and the **Gryfn WebUI**. The WebUI uses each sensor's *calibrated* focal length, whereas the calculator uses the manufacturer's nominal value (VNIR ~12.6 mm, RGB ~24 mm). Calibrated focal lengths are typically slightly shorter, marginally reducing the computed frame rate and overlap, but the difference is negligible (~0.01 ms in frame period, ~0.02% in sidelap) and is absorbed by the oversampling and sidelap margins. Either source is acceptable; the values in the table above are from the flight calculator.

Note

Type 1 (30 m) uses 20% oversampling, not the 30% used for Types 2–3. At 30 m the finer ground sampling requires a higher frame rate, so 30% oversampling pushes the VNIR frame rate close to the sensor's maximum, which GRYFN does not recommend. 20% oversampling keeps a safe margin below the maximum frame rate while still meeting the minimum oversampling requirement.

Onsite Preflight Operations

1. Conduct airspace and weather checks.

- No cloud cover.
- Maximum wind speed for quality hyperspectral capture is 6 m/s (22 km/h). Any wind speed over 5 m/s (18 km/h) should be recorded.
- Do not operate in conditions below 0 °C or above 40 °C.

- Try to ensure the sun is within $\pm 20^{\circ}$ of solar noon (approximately 2 hours before or after noon). **However**, *this depends on time of year and latitude — check NOAA solar calculator if unsure.*

2. Turn on the aeronautical radio and set to local CTAF (find in ERSA).
3. *Optional, if over 50 km from a CORS station:* set up the Emlid RTK (install a peg or permanent GCP below the base station for recurring flights), let it run for at least 15 minutes, and start recording the RINEX file before flying.
4. Set up reflectance panels, the validation panel, and GCPs following the Standard Flight Procedure. In summary:

- Use the Dual ELM panel flight layout by default (mandatory for missions $>$ 15 minutes).
- Fall back to the Single ELM panel flight layout only when a second panel set is unavailable or for short missions ($\leq$ 10 minutes) under stable illumination.
- Deploy the 2-panel validation set with its paired elevated + ground GCPs per Validation Panel Setup.
- Follow the Panel Setup rules (level folding table, panels in brightness order aligned with the flight direction, record panel-surface height).
- Place 5 GCPs per Ground Control Points — in alternate flight lines, next to the calibration panels.
- When flying multiple missions, panels and GCPs must be present in every flight (panels overflown approximately every 15 minutes).

Figure: Correctly deployed 4-panel ELM reflectance set on a levelled folding table. See the Standard Flight Procedure for the full panel and GCP layout diagrams.

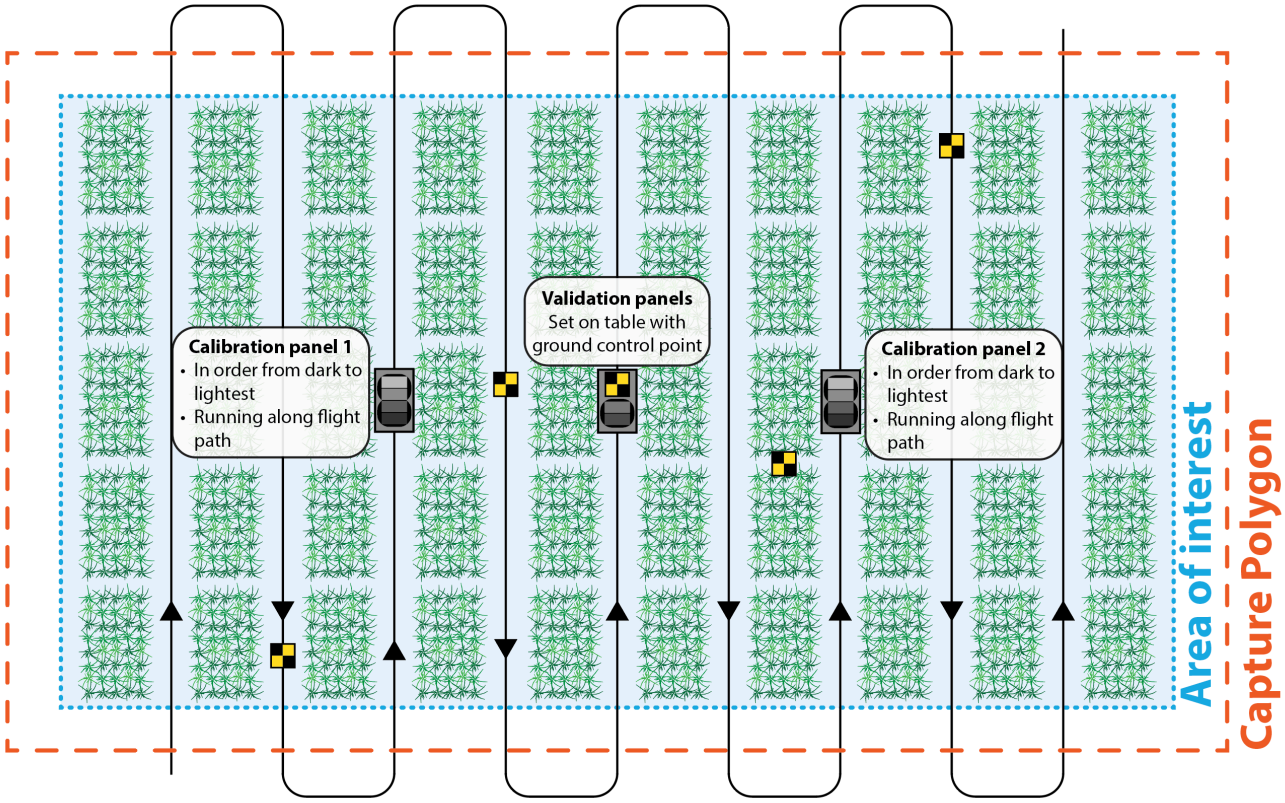

Figure: Dual ELM panel flight layout — default layout for GOBI M350 surveys. See the Dual ELM panel flight section of the Standard Flight Procedure for the full rationale and alternative layouts.

5. Set up the landing pad and UAV in a safe RTH location.

- In dusty environments an additional tarp should be used under the landing pad.

6. On the controller, set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings.
7. Attach the GOBI payload to the aircraft:

- Check that all lenses and sensors are clean. If not, use professional lens cleaning wipes to clean them.
- Connect the power cable from aircraft to payload (non-standard payload bus only).
- Connect both GNSS antenna cables to the correct ports (match A1 and A2).
- **Remove RGB and Nano HP lens caps.**
- Insert the RGB SD card.
- Connect the C-type power cable to power the gimbal (make sure the B side of the cable is toward the outer side of the drone and the upper side of the light sensor).
- *Optional, in warm ambient conditions:* activate the battery-powered fan. Avoid leaving GOBI powered on in stagnant air or high heat (> ~35–38 °C).

8. Power on the radio controller; check battery status.
9. Launch DJI Pilot 2 on the M350 controller.
10. Review the flight plan, checking operational height and double-checking that area of interest, GCPs, and reflectance panels are all within the capture polygon.
11. Power on the aircraft; confirm connection to RC/GCS and battery status.
12. If you are using RTK, check that you are receiving GNSS corrections. There should be more than 8 satellites for good correction (check the RTK fix in the M350 controller settings).

GOBI Sensor Configuration

1. Place the 80% exposure reference panel under the Nano HP lens; avoid casting shadows.

- Place the drone legs upon exposure-angle cones, ensuring a fixed angle is achieved each exposure setting.
- The maximum distance from the VNIR HS sensor to the GOBI reflectance panel is 40 cm. This is calculated based on the FOV and a 20 cm panel size.

2. Connect to GOBI Wi-Fi or connect the ethernet cable to the payload. *If using ethernet, ensure a static IP address for the sensor at* `10.0.65.2`.
3. Navigate to the GOBI WebUI at `10.0.65.50`.
4. Ensure a valid altitude is shown in the top-right corner of the WebUI (~10–20 m error is acceptable). (accurate position/elevation should be within ~30-60 seconds from power on)
5. Name the mission, using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
6. Ensure the correct UAV (M350) is selected in the dropdown box. **This is very important.**

> **Caution**
>
> Selecting the incorrect drone will render all data captured unrecoverable

7. Scroll down to polygon settings. Ensure triggering is set by **Polygon** in the dropdown menu. Set min and, optionally, max altitude triggers (recommended to set a minimum of at least 20 m to prevent unnecessary data capture).
8. Import the polygon KML or select from previously flown missions using the dropdown menu. Inspect the map to ensure it aligns with expectations.
9. Return to the top of the WebUI, open the Flight Calculator, and update flying height / flight speed (default 40 m height, default speed 3 m/s); adjust the oversampling buffer (recommended setting is 30%). Frame period will be automatically updated.
10. Adjust exposure until ~95% spectral saturation at the **peak of the 90th percentile line** without exceeding the frame period. Use the lowest gain mode that still provides sufficient saturation.
11. Place the lens cap on the Nano HP.
12. Click **Collect Dark Reference**; wait to finish.
13. **Remove the lens cap.**
14. Remove the exposure reference panel and return it to its case.
15. Press **Start Mission**.
16. Remove the ethernet cable (if using) and begin flight operations.

Flight Operations

1. Ensure the aircraft is in **Position** flight mode.
2. Clear people/objects away from the UAV.
3. Notify crew/observers that takeoff is beginning.
4. Begin manual takeoff, check stick controls work, and then fly to ~12 m AGL; **do not exceed** the trigger altitude (20 m default) before mission start.
5. Perform dynamic alignment (one to two figure-8 patterns) at a lower height than capture height and outside of the capture polygon area.
6. Enable autonomous mission.
7. Monitor UAV battery voltage/percentage in flight.
8. After mission, switch back to **Position** mode to regain manual control.
9. Lower to below capture height and perform post-mission dynamic alignment (figure-8).
10. Land and disarm UAV. Be careful when landing the M350 — downward sensors are blocked when using GOBI, so bring it in slowly.
11. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks. Ensure data transfer is finished before turning off the sensor.

> **Warning**
>
> **There is no hot-swap protection on the GOBI.** Do not mate or de-mate the Smart Dovetail while the aircraft is powered.

Warning

Hot-swapping batteries is **not recommended** with any high-end GNSS solution. Static data can cause IMU drift and degrade post-processed trajectory quality. **Treat each flight as a new flight.**

Post-Flight Sensor Configuration

1. Reconnect to GOBI Wi-Fi or ethernet cable.
2. Open a browser to `10.0.65.50` or refresh the UI.
3. Press **Stop Mission**.
4. Replace the Nano HP and RGB lens caps.
5. In the WebUI, ensure all data has captured correctly (see the Data Confirmation table below).

Data Confirmation

Sensor type	File count	File size	Notes
GNSS-INS	1 file	~1.5 MB per minute	A new file will be created if time rolls over the hour (UTC time, not capture time).
LiDAR	1 per minute	~175 MB per file	First & last file may be smaller. Files may be smaller if over a low-reflectivity object.
VNIR	Mission time (s) / Frame Period value ~ number of data cubes	Several GB even for short flights.	Check that `frameindex`, `imu_gps`, and `settings` files all exist.
RGB	One image every 2 s* (by default)	~30–70 MB per image	Check the event file vs the number of images.

Post-Flight

1. Power off the drone, then the controller. Inspect the drone for any damage and report for maintenance if necessary.
2. Disconnect the ethernet cable from the payload, if using.
3. Disconnect the aircraft batteries.
4. Disconnect the payload power cable.
5. Disconnect the payload GNSS cables.
6. Undo the mounting clamp, remove the payload, and place it in its case.
7. Pack up the drone and all gear from site, double-checking against the checklist at the start of this document to ensure everything is accounted for.

Offload Data

1. Open WinSCP, FileZilla, or similar FTP software.
2. Connect to the GRYFN Gobi via `gryfn@10.0.65.50` (password: `gryfn`).
3. Raw GNSS & LiDAR PCAP location: `/data/{mission name}`.
4. VNIR data location: `/data/capturedData/captured/{dateTime}`.
5. GOBI logs location: `/data/gryfn.log.{date}`.
6. RGB images are stored on the camera SD card.
7. Download GNSS, LiDAR, logs, and RGB after each mission; download hyperspectral data later due to its size.
8. Clear data directories after download and backup to avoid filling the 500 GB SSD.

- Gryfn Processing Tool v1.94 and greater can query the sensor directly with the new firmware and so can be used to offload the data. In some cases where the drone is powered off too quickly after stopping the mission the GNSS data may be absent. This can be recovered by connecting directly to the SBG via FTP — instructions for connection below.

Data storage, processing & validation

All paths below follow the APPN folder structure. Formal paths use the wiki's placeholder syntax; an example follows each.

1. Downloaded data should be stored in the correct `T0_raw` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/
```

2. Data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/Vault/
```

3. Important information and issues should be recorded in the `FieldNotes.txt` file.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/FieldNotes.txt
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/FieldNotes.txt
```

4. Per-run information (e.g. APEx test conditions) should be stored in `RunOverview.csv`.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/RunOverview.csv
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/RunOverview.csv
```

> **Important**
>
> When data from a failed run is being kept (e.g. debugging with GRYFN), the `RunFailed` boolean column of `RunOverview.csv` must be set to `True`.

5. The bundled `.graw` should be saved in the same `T0_raw` folder.
6. The `.gpro` should be generated using the APPN GPT Pipeline (see the Processing Pipelines document and the `Gobi_standard_pipeline_v1.0.yaml` in the adjacent `yaml/` folder) and stored in the adjacent `T1_proc` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T1_proc/
```

7. Standard QA process should be performed following this guide.

> **Note**
>
> Missing pixels along VNIR image edges can result from insufficient overlap, trajectory irregularities, overly restrictive KML boundaries, or a restrictive processing extent. Where practical, process with no fixed extent (or against a generously buffered capture polygon) so valid edge data is not clipped.

Appendix

FTP & Service Reference

Service	Protocol	IP	Port	User	Password
GOBI	FTP	`10.0.65.50`	22	`gryfn`	`gryfn`
SBG	FTP	`10.0.65.100`	21	`operator`	*(none)*
Headwall polygon	HTTP	`apps.headwallphotonics.com`	—	—	—

Headwall Polygon Tool — Workflow

1. Import KML.

- Must be in Google Earth format.

2. Export KML.
3. Save KML.
4. In File Explorer, rename the KML as the survey name.

Setting a Static IP Address

If connecting over an ethernet connection, a static IP address will need to be set:

1. Open *Network and Internet Settings*.
2. Open *Change Adapter Options*.
3. Open *Properties* for the Ethernet port/adapter.
4. Open *Properties* for IPv4.
5. Set IP address to `10.0.65.2`.
6. Set subnet mask to `255.255.255.0`.
7. Press **OK**.

IP Address Reference

Service	Address
GRYFN WebUI	`10.0.65.50`
HSInsight	`10.0.65.50:8080`
Ouster	`10.0.65.128`
SBG	`10.0.65.100`